

Config-Driven RL for Just-in-Time Adaptive Interventions

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Problem Definition

Goal: Optimise personalised physical-activity interventions with reinforcement learning.

MDP formulation: $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$

- $\mathcal{S}$: 36-state factored MDP (step bin, sleep quality, day of week, burden)
- $\mathcal{A}$: 4 actions (idle, movement suggestion, goal reminder, journal)
- $\mathcal{P}$: Unknown transition dynamics, estimated from LLM simulation or curated tables
- $\mathcal{R}$: $\alpha \cdot f(\text{step_bin}) + (1-\alpha) \cdot g(\text{sleep}) - \lambda \cdot \mathbb{I}[\text{action} \neq \text{idle}]$

Core challenge: How do we iterate on RL agents without live patients?

Approach: Use a config-driven simulator with LLM-bootstrapped user models.

Existing Work

	HeartSteps V2	PEARL	StepCountJITAI
Type	MRT	4-arm RCT	Simulator
Sample	$n \approx 97$	$n = 13,463$	Synthetic
Algorithm	TS bandit	RL bandit	DQN
Actions	~ 5 types	155 nudges	4 types
Outcome	78.4% improved	+296 steps*	DQN ≈ 3000
Theory	SDT / FBM	COM-B	—

*RL vs control at 1 month, $p = 0.0002$.

Key gap: No existing framework combines *config-driven experiment definition* with *LLM-bootstrapped user simulation* for rapid agent prototyping before real-world deployment.

Our Approach

Two pillars:

① Config-Driven Framework

- Entire MDP defined in YAML, with no source-code changes
- 36-state factored environment, expression-based rewards, 3 transition types
- 7 RL/bandit agents, evaluation runner with multi-seed benchmarking

② Synthetic LLM Patients

- 22,320 prompts per persona across 5 behavioural personas
- LLM-as-transition-model samples next-state probabilities
- Total API cost: **\$0.27** for the entire bootstrap

What We Have Done

	MVP Simulator	LLM Bootstrap	Experiments
Purpose	Prove the pipeline	Generate patient behaviour	Evaluate agents
State space	2-state	36-state factored	Both
Transitions	Hand-curated rules	22,320 LLM prompts	YAML-defined
Output	285 configs	42 tables, 85 figures	7 personas $\times$ 7 agents
Cost	\$0	\$0.27 total	\$0

Key point: The 2-state curated MDP and 36-state LLM-bootstrapped MDP use the *same* YAML schema, agent library, and evaluation runner.

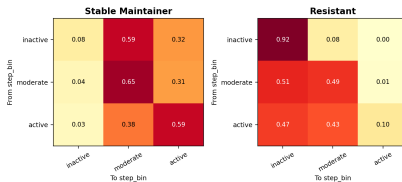
Initial Results

Transition analysis

- Personas separate clearly under the same intervention.
- Notification burden reduces movement-suggestion effectiveness.
- Learned transition tables encode plausible behavioural heterogeneity.
- HeartSteps V2 agent outperforms all bandit baselines.

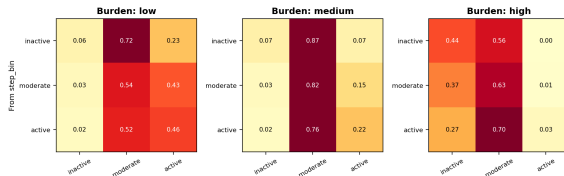
Persona contrast: same action, different transition patterns.

Persona-Specific Transition Patterns Under Movement Suggestion
Averaged across all conditions



Burden effect: higher notification burden weakens suggestions.

Movement Suggestion Effectiveness Degrades Under Higher Burden
Base Persona, averaged across time-of-day, day boundary, sleep



Next Steps

① Broader Agent Suite

- Add PEARL-style contextual bandit/RL agents for direct comparison
- Stress-test offline RL, constrained RL, and adaptive bandit baselines

② Deep Learning Agents

- Benchmark DQN, actor-critic, and sequence-based policies
- Explore representation learning for high-dimensional wearable state inputs

③ More Complex Environments

- Expand state spaces with context, history, fatigue, and adherence dynamics
- Model non-stationarity, delayed rewards, and intervention burden

④ Sensor Foundation Models

- Wearable foundation-model embeddings for dynamic action spaces
- Connect interventions to sensor-derived activity and sleep context

⑤ LLM and Prompt Variants

- Compare DeepSeek, GLM, Qwen, Claude, and smaller local models
- Test COM-B prompts, structured JSON, and multi-turn persona dialogues